GDTCR: guide for supervisor review

Scientific draft prepared for discussion before journal submission

Purpose and positioning. The manuscript evaluates recovery of publicly recorded human gamma/delta TCR partners from one CDR3. Its contribution is a controlled benchmark separating sequence priors, conditional information, free candidate availability, ranking and exposure. BMC Bioinformatics is the primary preparation target; the accompanying journal comparison assesses two alternatives.

Central finding. Across the primary component-disjoint splits, conditional NLL is 1.005 nats per target token, compared with 1.013 for the matched prior and 1.115 for random initialization. However, primary likelihood Hit@5 at 64 draws is only 0.03%. Separate ranking does not show a reproducible primary improvement. Higher recovery in exposed-target populations coexists with substantial prior performance and concentrated events. Recorded recovery is not evidence of receptor function.

Reading order. Start with the structured abstract, Figure 1, and Results 3.1–3.3 with Tables 1–3. Figures 4–6 then distinguish constructed retrieval from free generation and explain graph-composition/exposure sensitivity. Discussion 4.1–4.4 states the interpretation and limits. Additional file 1 provides full methods, sensitivity analyses and Figures S1–S3; Additional files 2–6 retain numerical and reproducibility evidence.

Questions for the supervisor.

1. Is the benchmark question and contribution sufficiently clear for BMC Bioinformatics, and is the proposed title appropriate?

2. Are the controls, uncertainty units and interpretation of sparse hits convincing, especially in relation to target priors and sequence neighborhoods?

3. Are the source concentration, distant-pair scarcity and lack of donor-independent validation adequately bounded?

4. Is further evidence needed before submission, or should the current bounded benchmark be submitted first? Would a higher-positioned journal justify a separate study extension?

What changed. The draft consolidates the Results into six sections, reorganizes Methods and Discussion, and moves sampling/calibration detail to the supplement. Scientific data, fitted results, intervals and model settings remain frozen. Author details and administrative statements are outside these review PDFs. The separate submission checklist records decisions to complete after review.

Materials. Main and supplementary PDFs, editable LaTeX, six numbered attachments, separate figure assets and a consistency/placement audit accompany the draft. The public scientific base is https://github.com/he2238751697-afk/GDTCR.